

UDC

Valeriia Yahlo

higher education student, second (master's) level of higher education

ERI “Karazin Banking Institute”

V.N. Karazin Kharkiv National University

*Scientific advisor - **Victoria Shevchenko***

Candidate of Philological Sciences, Associate Professor

of the Department of Management, Accounting and Business Communications

ERI “Karazin Banking Institute”

of V.N. Karazin Kharkiv National University

DEVELOPING A METHOD TO COMBAT SOCIAL ENGINEERING AS A SOCIAL INNOVATION

In today's digital environment, where information technology has become an integral part of everyday life, the issue of information security is of particular importance. Almost all areas of human activity, including finance, communication, education, healthcare, and public administration, are closely connected with the use of digital technologies, which creates not only new opportunities but also significant risks related to data protection. As technical security measures continue to evolve, cyberattack methods are becoming increasingly sophisticated, adapting to new conditions and exploiting system vulnerabilities. At the same time, the human factor remains one of the weakest elements in any security system, which makes it a primary target for attackers.

Social engineering represents a set of psychological manipulation techniques aimed at influencing individuals in order to obtain confidential information, gain unauthorized access, or induce specific actions. Unlike purely technical attacks, social engineering relies on trust, emotions, lack of awareness, or inattention of users. Attacks based on social engineering methods often do not require advanced technical skills or significant resources, yet they can cause substantial harm to both individuals and organizations. Common techniques include phishing, vishing, smishing, identity theft, and the creation of fake websites, all of which are widely used to steal personal data, financial credentials, and access to corporate systems. These threats are particularly critical in sectors such as banking, education, government, and healthcare, where large volumes of sensitive information are processed.

The rapid growth of remote work, online learning, and digital services further increases the exposure of users to such risks, especially when they lack sufficient

knowledge and practical skills in cybersecurity. Recent analyses indicate a significant rise in incidents related to social engineering, highlighting the urgent need for effective preventive measures. Despite the general increase in digital literacy, many users still do not possess basic skills to recognize manipulation, detect fraudulent activities, or respond appropriately to suspicious situations. Moreover, successful social engineering attacks can lead not only to financial losses but also to data breaches, reputational damage, disruption of organizational processes, and even threats to national security. In this context, raising user awareness and developing practical cybersecurity skills becomes a critical priority.

One effective solution to this problem is the creation of specialized educational resources that combine theoretical knowledge with practical training. The proposed approach involves the development of an interactive educational website aimed at counteracting social engineering. The primary goal of this resource is not only to inform users about existing threats but also to develop устойчивые behavioral patterns that enable them to resist manipulation and make safe decisions in real-life situations. The concept of the website includes a structured division into several functional sections. The theoretical section provides comprehensive and systematized information about the main types of social engineering attacks, their characteristics, common scenarios, and prevention strategies, presented in an accessible format with real-life examples to enhance understanding. The practical section is implemented through interactive tests, simulations, and scenario-based tasks that replicate typical fraudulent situations, allowing users to make decisions in a safe environment and receive immediate feedback with explanations of correct and incorrect actions.

This approach not only knowledge retention but also the development of critical thinking and analytical skills. Additionally, the platform may include concise guidelines and summaries for quick reference, as well as analytical components such as user surveys to assess awareness levels and identify knowledge gaps. Special attention is given to usability, ensuring that the website interface is intuitive, user-friendly, and adaptable to different age groups and levels of digital literacy. The resource can be used both for self-learning and as part of formal educational programs, corporate training, or institutional awareness campaigns. Its integration into the activities of banks, educational institutions, and other organizations can significantly improve the overall level of information security among employees and clients. The practical value of the developed website lies in its ability to directly influence the level of digital security in society by combining theory and practice, enabling users to gain not only knowledge but also hands-on experience in recognizing and responding to threats. This significantly reduces the likelihood of successful attacks and promotes responsible behavior in the digital space. The implementation of such a resource will contribute to reducing fraud incidents,

financial losses, and data breaches, while also enhancing public trust in digital services and supporting the development of a secure digital economy.

Therefore, the development of an interactive educational website to counter social engineering is a timely and socially significant initiative that combines an innovative educational approach with practical benefits for a wide range of users and provides an effective response to modern cybersecurity challenges.

RESOURCES

1. Mitnick K., Simon W. The Art of Deception: Controlling the Human Element of Security. — Wiley, 2002.
2. Hadnagy C. Social Engineering: The Science of Human Hacking. — Wiley, 2018.
3. National Institute of Standards and Technology (NIST). Guide to Enterprise Telework, Remote Access, and Bring Your Own Device (BYOD) Security.
4. ENISA (European Union Agency for Cybersecurity). Cybersecurity Awareness Raising Guide.
5. Verizon. Data Breach Investigations Report